

Title

Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support

Abstract

This interim status report presents the detailed progress of the ongoing research project focused on developing a multimodal artificial intelligence system for skin disease diagnosis. The proposed system integrates multiple data types including dermatological images, patient-reported symptoms in text and audio formats, and clinical information to improve diagnostic accuracy and reliability. The research addresses key limitations in existing AI systems such as lack of multimodal integration, bias across skin tones, and absence of personalized treatment support. The system aims to provide accurate predictions, fairness across populations, and explainable outputs for both patients and healthcare professionals. The expected outcome is a scalable, user-friendly, and efficient decision-support system.

Introduction

Skin diseases are among the most common medical conditions worldwide. Early diagnosis is critical to prevent complications and ensure effective treatment. However, access to dermatologists is limited, especially in developing regions. Traditional AI systems mainly rely on image-based classification, which reduces accuracy when diseases share similar visual patterns. This research introduces a multimodal approach that combines images, text, and audio inputs to improve diagnostic performance and usability.

Objectives

The main objective of this research is to develop an accurate, fair, and personalized multimodal AI system for skin disease diagnosis and treatment support. Specific objectives include developing image-based classification models, implementing symptom analysis, integrating multimodal fusion techniques, improving fairness across skin tones, monitoring disease progression, and generating personalized treatment recommendations.

Literature Review

Previous research shows that deep learning models such as CNNs can achieve high accuracy in skin disease classification. However, most systems rely only on image data. Recent studies highlight the importance of multimodal learning, combining images with text or metadata to improve performance. Despite advancements, challenges such as bias, lack of personalization, and limited real-world applicability remain unresolved.

Methodology

The system follows a structured approach including data collection, preprocessing, model development, and integration. CNN models are used for image analysis, while NLP models process textual symptoms. Audio inputs are converted into text using speech recognition. Multimodal fusion combines outputs from different models to generate final predictions. Domain adaptation techniques are used to reduce bias and improve fairness.

Work Completed

The literature review has been completed, and key research gaps have been identified. Initial datasets have been collected and preprocessed. Image-based classification models have been implemented and tested. Text processing modules have been developed using NLP techniques. Preliminary experiments show improved performance when combining multiple data sources.

Results So Far

Initial results indicate that multimodal approaches outperform image-only models. The system achieves higher Top-3 accuracy, which is important in medical diagnosis. Early testing shows that integrating symptom data improves prediction reliability.

Challenges

The main challenges include data imbalance across skin tones, limited availability of audio datasets, complexity in integrating multiple models, and ensuring system scalability. Addressing these challenges is essential for building a robust system.

Future Work

Future work includes improving model performance, expanding datasets, implementing full multimodal integration, developing a user interface, and conducting system evaluation. Additional work will focus on deployment and real-world testing.

Conclusion

The project is progressing successfully and demonstrates strong potential in improving skin disease diagnosis using AI. The proposed multimodal framework addresses key limitations in existing systems and aims to provide a practical healthcare solution.

Additional Discussion 1

This section provides extended discussion on system design, implementation challenges, and expected impact. The multimodal framework is designed to improve healthcare accessibility, particularly in regions with limited medical resources. By integrating advanced AI techniques, the system aims to support both patients and healthcare professionals in making informed decisions. Further development will focus on optimizing model performance, enhancing user experience, and ensuring ethical AI practices.

Additional Discussion 2

This section provides extended discussion on system design, implementation challenges, and expected impact. The multimodal framework is designed to improve healthcare accessibility, particularly in regions with limited medical resources. By integrating advanced AI techniques, the system aims to support both patients and healthcare professionals in making informed decisions. Further development will focus on optimizing model performance, enhancing user experience, and ensuring ethical AI practices.